第二十讲
稳定概念、欧拉公式

作业：11-1,2,4,5,7

第十一章　压杆稳定

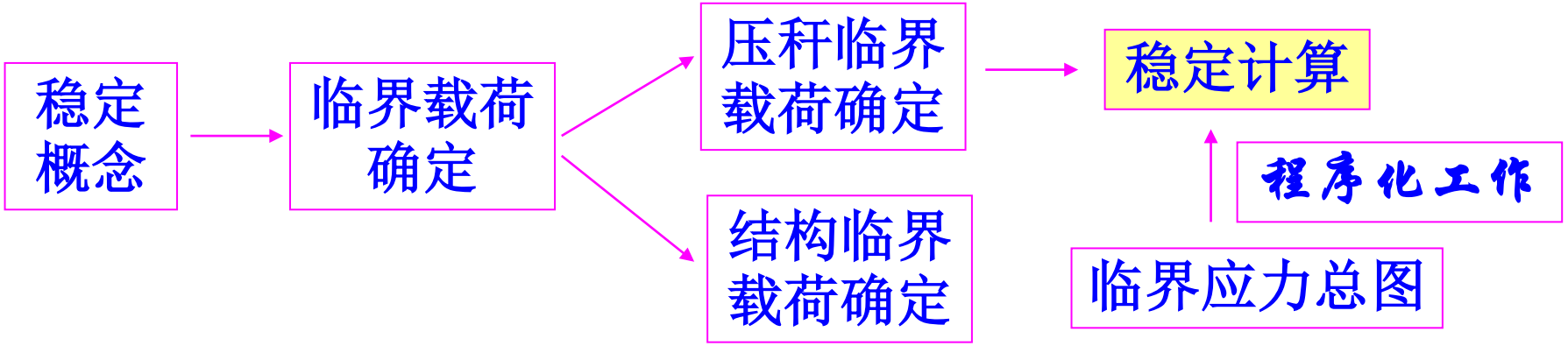
11.1 稳定概念

11.2 两端铰支细长压杆的临界载荷

11.3 两端非铰支细长压杆的临界载荷

11.4 中小柔度杆的临界应力——临界应力总图

11.5 压杆稳定条件与合理设计



第十一章 压杆稳定

11.1 稳定概念

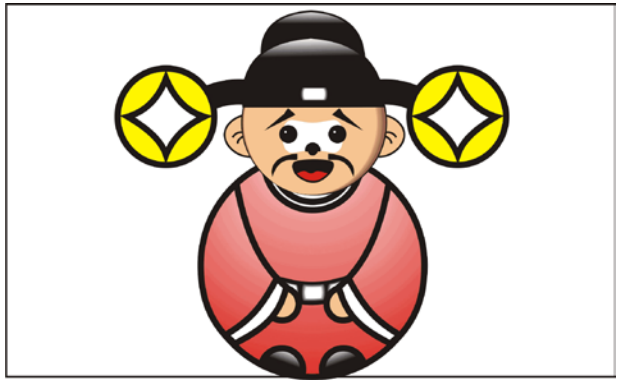
1、平衡状态的稳定性

2、压杆稳定概念

3、结构的稳定问题

4、稳定问题研究目的与方法

什么是稳定？举例



控制系统；
人的情绪、心态；
人际、国际关系；
社会秩序；
……

与扰动导致偏离相关

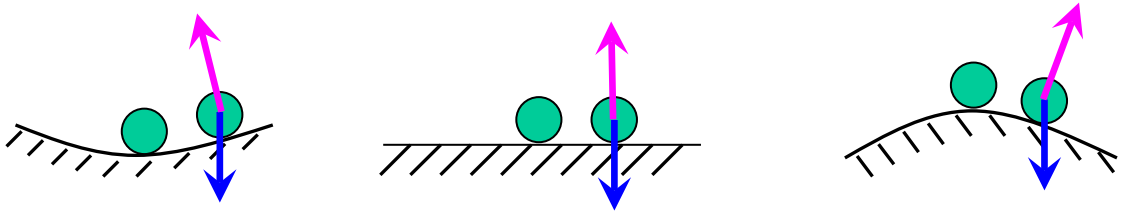
1、平衡状态的稳定性

存在扰动

随机的小载荷，风、振动等

小球三种平衡状态

稳定吗？



取决于支撑面构型

平衡状态 —扰动→ 微小偏离 邻近状态 —扰动→

- 恢复 ——稳定平衡
- 继续偏离——不稳定平衡
- 不变 ——临界平衡

稳定性：维持原有平衡状态的能力

稳定 ≠ 恒定

扰动导致偏离

分析邻近状态的受力

不经历风雨，怎么见彩虹？

患难见真情

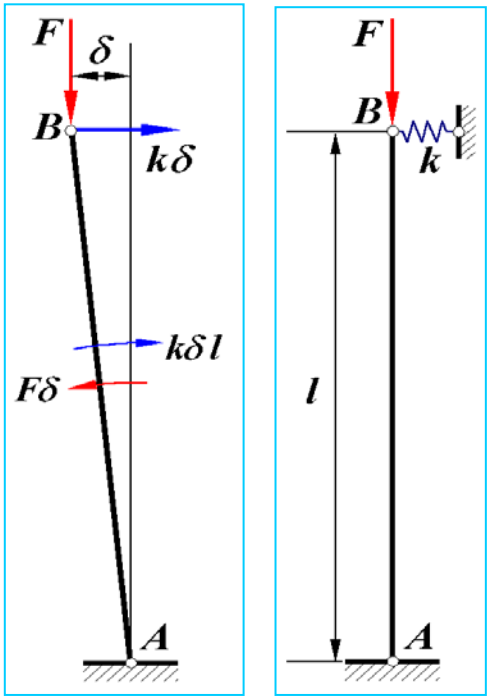
1、平衡状态的稳定性

刚杆–弹簧系统的稳定性

直线平衡状态 → 扰动 → 微小偏离δ 邻近状态

$F\delta$ - 偏离力矩
$k\delta l$ - 恢复力矩

$F < kl$ 恢复——稳定平衡；
$F > kl$ 偏离——不稳定平衡（失稳）；
$F = kl$ 不变——临界平衡

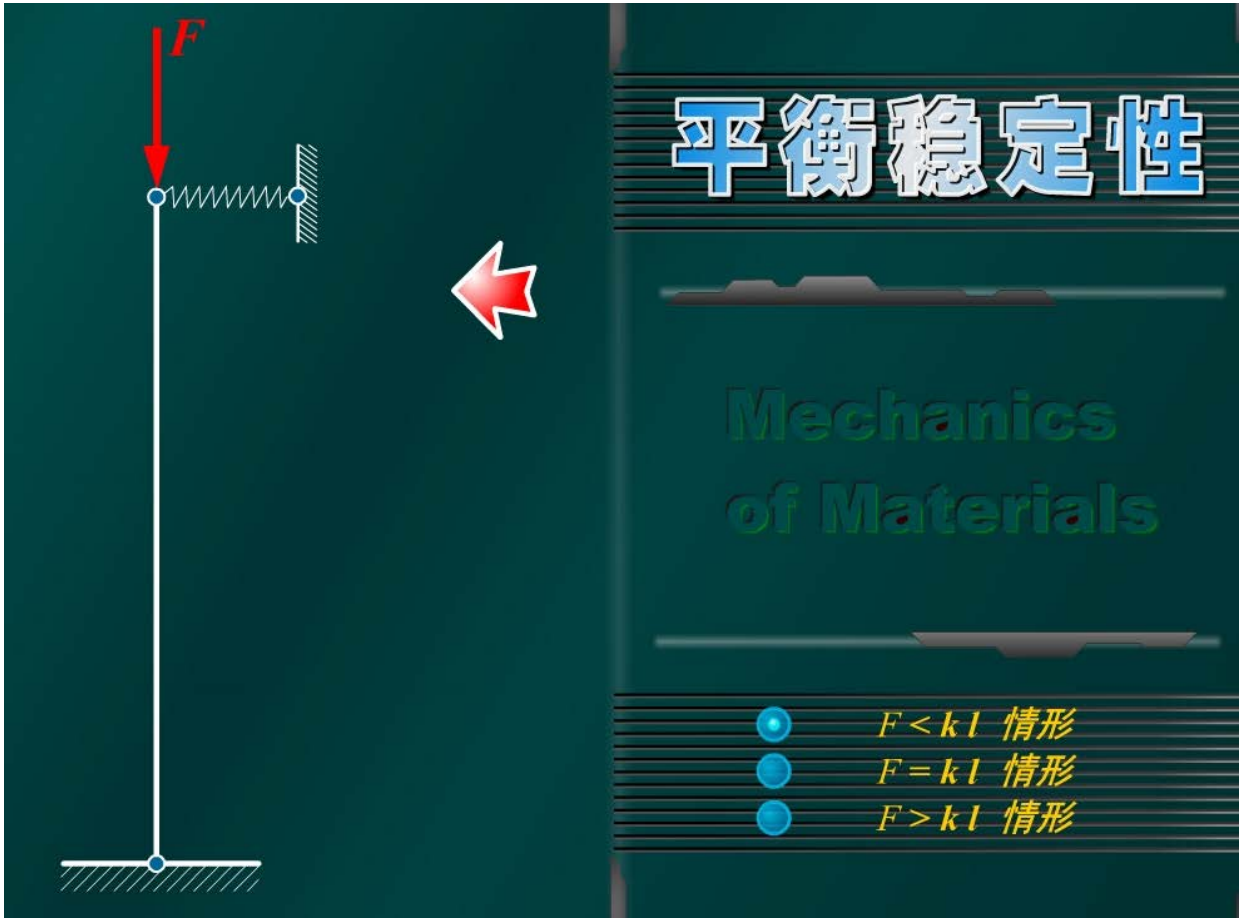


临界载荷 $F_{cr} = kl$

- 使系统在邻近状态平衡的载荷；
- 系统由稳定平衡到不稳定平衡的临界载荷；
- 是系统抗失稳的能力，取决于系统特性（弹簧刚度和杆长）。

1、平衡状态的稳定性

刚杆–弹簧系统的稳定性



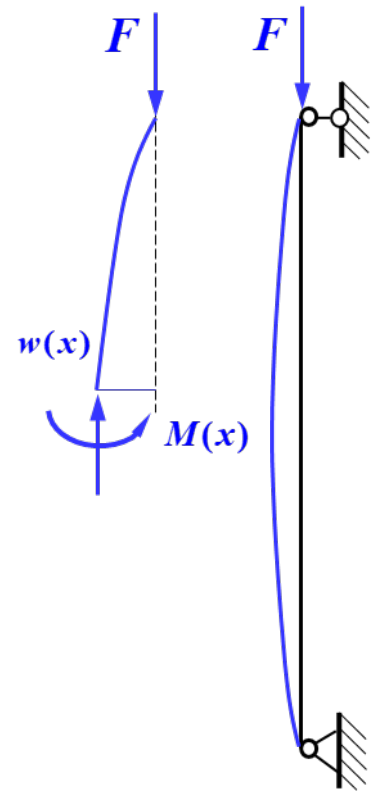
2、压杆稳定概念

细长理想直杆轴向受压

$$\frac{d^2w}{dx^2}=\frac{M(x)}{EI}$$

直线平衡 → 扰动 → 微弯$w(x)$ 邻近状态

弯矩与挠度相关



$Fw(x) < M(x)$ 恢复—稳定平衡；

$Fw(x) > M(x)$ 偏离—不稳定平衡（失稳）；

$Fw(x) = M(x)$ 不变—临界平衡（微弯平衡）

$F < F_{cr}$

$F > F_{cr}$

$F = F_{cr}$

临界载荷 F_{cr} 失稳（屈曲）buckling

- 使压杆在微弯状态保持平衡的轴向压力；
- 压杆稳定与失稳的临界轴向压力；
- 是压杆抗失稳的能力，取决于压杆的弯曲刚度（偏离直线平衡的难易程度）。

$Fw(x)$ - 偏离力矩

$M(x)$ - 恢复力矩

2、压杆稳定概念

细长理想直杆轴向受压

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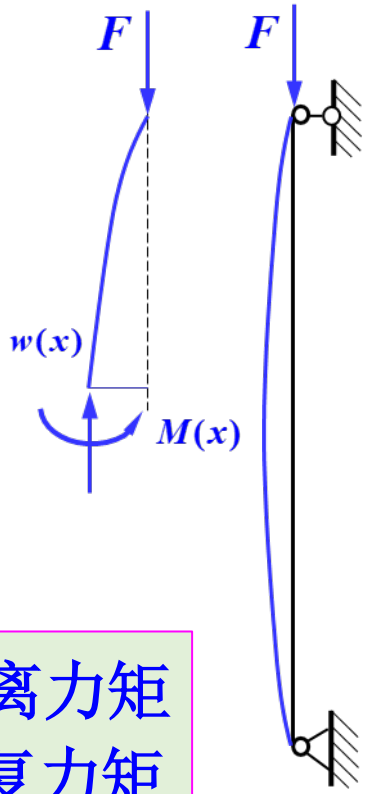
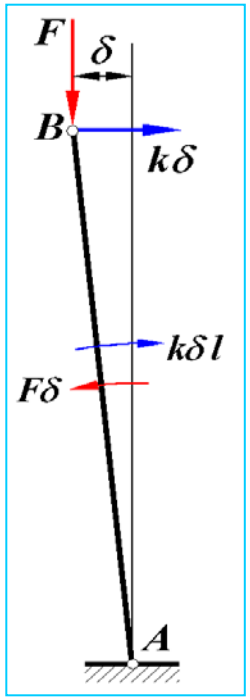
$$\frac{d^2w}{dx^2}=\frac{M(x)}{EI}$$

直线平衡 → 扰动 → 微弯$w(x)$ 邻近状态

弯矩与挠度相关





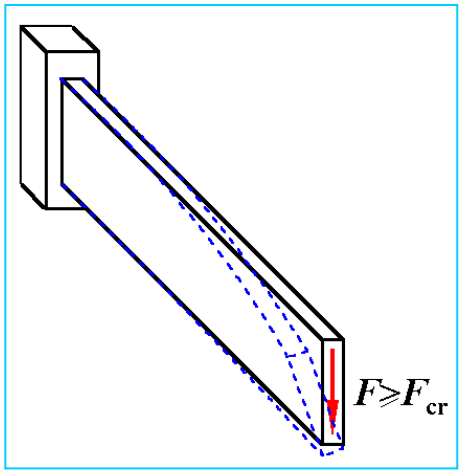
如何理解恢复力矩

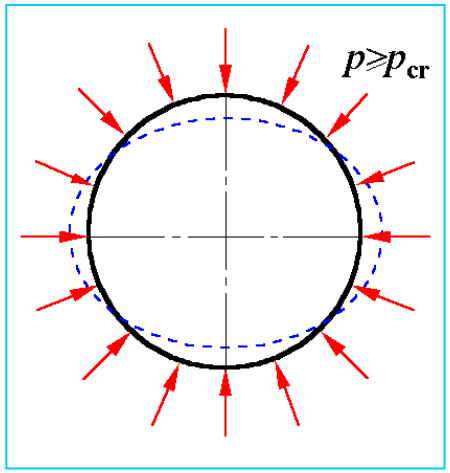
$F\delta$ - 偏离力矩
$k\delta l$ - 恢复力矩

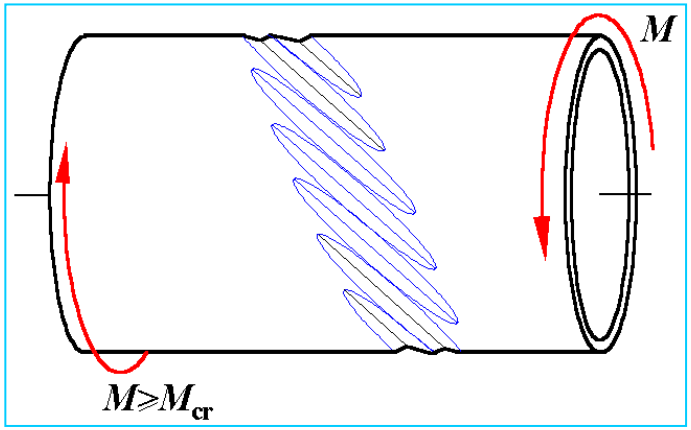
$Fw(x)$ - 偏离力矩
$M(x)$ - 恢复力矩

弹性体具有恢复变形的能力 → 构件内力；
弯矩是内力，与曲率相关，表征恢复能力。

3、结构的稳定问题







4、稳定问题研究目的与方法

目的： 求临界载荷F_{cr}，保证结构不失稳

方法：

1、一般而言，使结构偏离原平衡状态（处于邻近状态），维持其平衡的外载荷就是临界载荷；

2、对压杆而言，利用微弯状态下的平衡条件求临界载荷

第十一章 压杆稳定

11.2 两端铰支细长压杆的临界载荷

1、临界载荷的欧拉公式推导

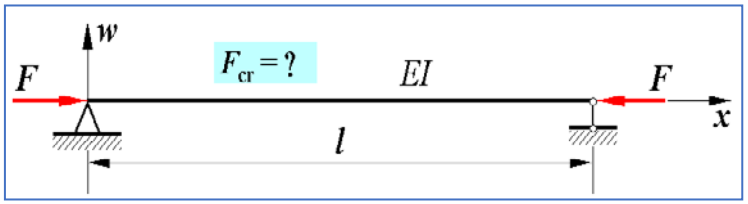
2、关于临界载荷的讨论

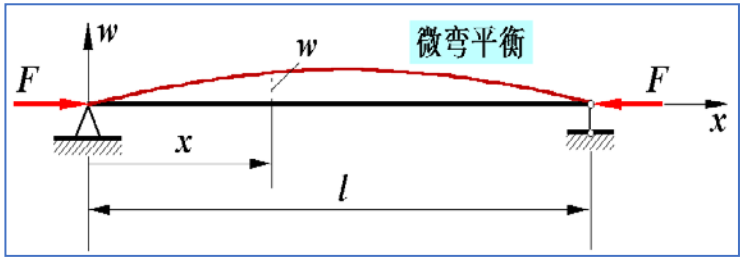
3、例题

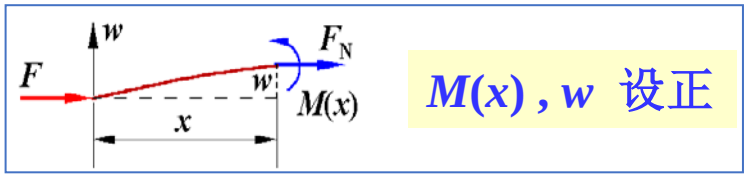
1、临界载荷的欧拉公式推导

两端铰支细长理想直杆受压

求解方法：利用微弯平衡







$$M(x)=-Fw \qquad \frac{d^2w}{dx^2}=\frac{M(x)}{EI}$$

$$\frac{d^2w}{dx^2}+k^2w=0 \qquad \left(k^2=\frac{F}{EI}\right)$$

- 以挠度表示的微弯平衡方程；
- 关于挠度的齐次微分方程。

二阶微分方程边值问题的特征值与特征函数

$$w=A\sin kx+B\cos kx$$

位移边界条件：

$x=0,\ w=0$ → $B=0$

$x=l,\ w=0$ → $A\sin kl=0$

解答的非零解条件 → $\sin kl=0$

$$kl=n\pi, \qquad F=\frac{n^2\pi^2EI}{l^2}, \quad (n=0,1,2,\cdots)$$

$$F_{cr}=\frac{\pi^2EI}{l^2}$$

欧拉公式

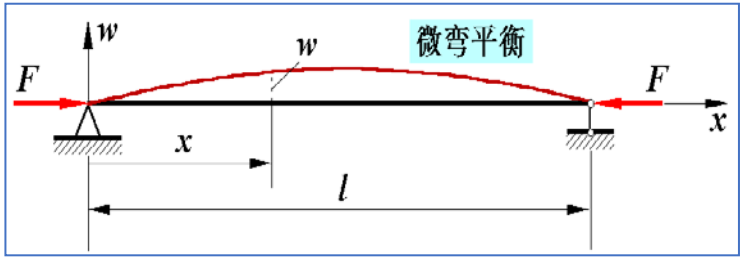
欧拉临界载荷

1、临界载荷的欧拉公式推导

讨论

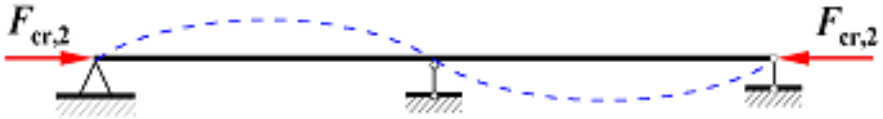
$$F=\frac{n^2\pi^2EI}{l^2},\quad (n=0,1,2,\cdots)$$

$$w=A\sin\frac{n\pi}{l}x$$



$$F_{cr}=\frac{\pi^2EI}{l^2}$$

$$w=A\sin\frac{\pi x}{l}$$



1）$F_{cr}\propto EI,\quad F_{cr}\propto 1/l^2$

2）在最小刚度平面内失稳；

3）挠曲轴为半个正弦波——一阶失稳模态，幅值不定；

4）高阶临界载荷与高阶失稳模态：

$$n=2,\ \Rightarrow\ F_{cr}=\frac{4\pi^2EI}{l^2}=\frac{\pi^2EI}{\left(l/2\right)^2},\quad w=A\sin(\frac{2\pi}{l}x)$$

2、关于临界载荷的讨论

1）欧拉公式基于挠曲轴近似微分方程，F_{cr}作用下，微弯幅值不定，处于随遇平衡。只适用于小变形情况。

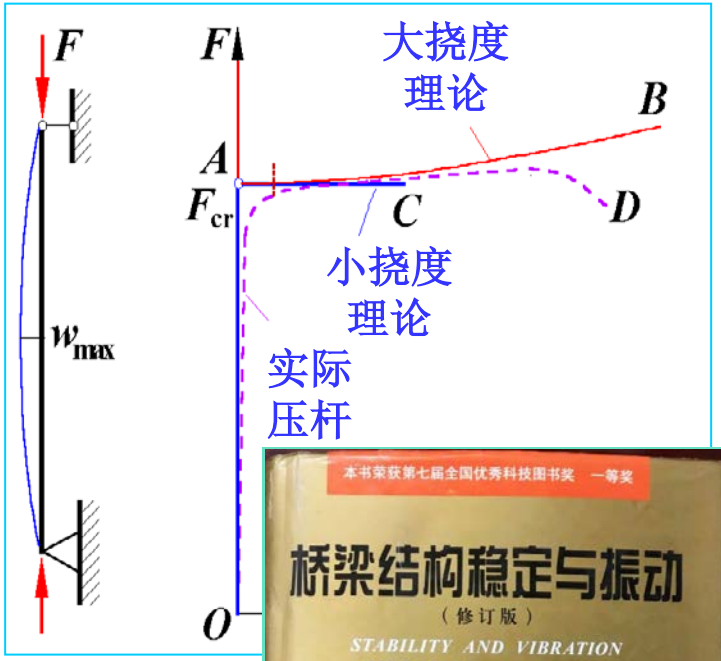
2）按照大挠度理论，不存在随遇平衡，轴压增大，横向挠度增加：

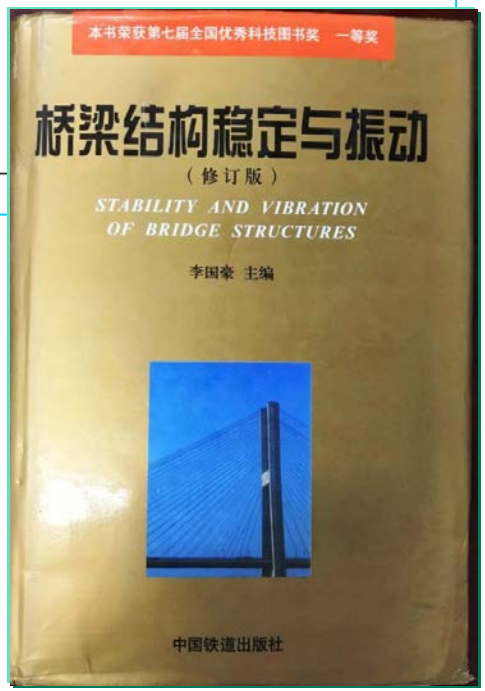
A、$F=F_{cr}$后，轴压微小增加，挠度急剧增大，表明F_{cr}反映了压杆抗失稳能力；

B、挠度增大之后，载荷增加表明，失稳后压杆仍有承载能力。

C、板、壳、加筋结构的后屈曲承载。

3）对于实际压杆（初始弯曲、轴压偏心、横向受载），轴压接近F_{cr}，挠度急剧增大，F_{cr}仍可用。

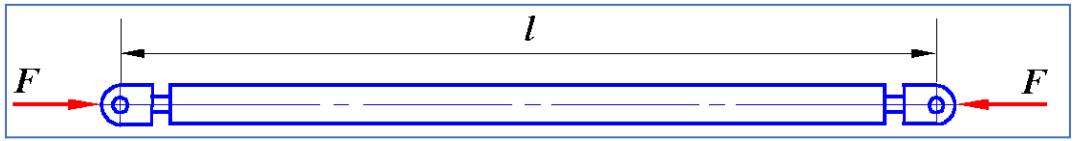




3、例题

例11-1 图示细长压杆， $l = 0.8$ m, $d = 20$ mm, $E = 200$ GPa, $\sigma_s = 235$ MPa, 求F_{cr}.

解：



$$F_{cr} = \frac{\pi^2 E}{l^2} \cdot \frac{\pi d^4}{64} = 24.2\,\text{kN}$$

$$F_s = \sigma_s \cdot \frac{\pi d^2}{4} = 73.8\,\text{kN} \qquad F_{cr} \ll F_s$$

细长杆的承压能力，是由稳定性要求确定的

第十一章 压杆稳定

11.3 两端非铰支细长压杆的临界载荷

1、一端铰支一端固定细长压杆

2、欧拉公式的一般形式

3、例题

微分方程解法

微弯平衡条件 $\frac{d^2w}{dx^2}+\frac{F}{EI}w(x)=R(x)$

位移边界条件

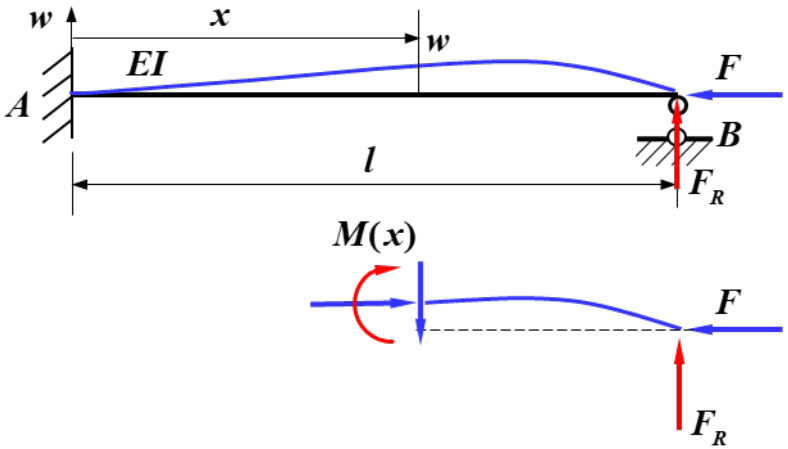
挠度的非零解条件

类比法

两端非铰支 ⟷ 两端铰支

微弯一致，轴压相同

1、一端铰支一端固定细长压杆



微弯平衡方程

$$M(x) = -Fw + F_R(l - x)$$

$$\frac{d^2w}{dx^2} + k^2 w = \frac{F_R}{EI}(l - x) \qquad \left(k^2 = \frac{F}{EI}\right)$$

挠度表达式

$$w = A\sin kx + B\cos kx + \frac{F_R}{EIk^2}(l - x)$$

位移边界条件

$$\begin{aligned} &x = 0, \quad w = 0 \\ &x = 0, \quad w' = 0 \\ &x = l, \quad w = 0 \end{aligned}$$

$$\Rightarrow$$

$$\begin{aligned} &B + \frac{F_R l}{EIk^2} = 0 \\ &Ak - \frac{F_R}{EIk^2} = 0 \\ &A\sin kl + B\cos kl = 0 \end{aligned}$$

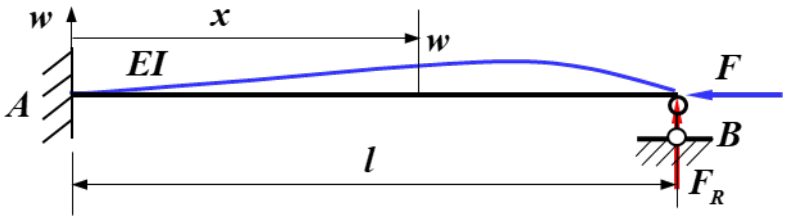
非零解条件

$$\begin{vmatrix} 0 & 1 & \frac{l}{EIk^2} \\ k & 0 & -\frac{1}{EIk^2} \\ \sin kl & \cos kl & 0 \end{vmatrix} = 0$$

$$\tan kl = kl$$

1、一端铰支一端固定细长压杆

临界载荷解答



$$\tan kl = kl$$

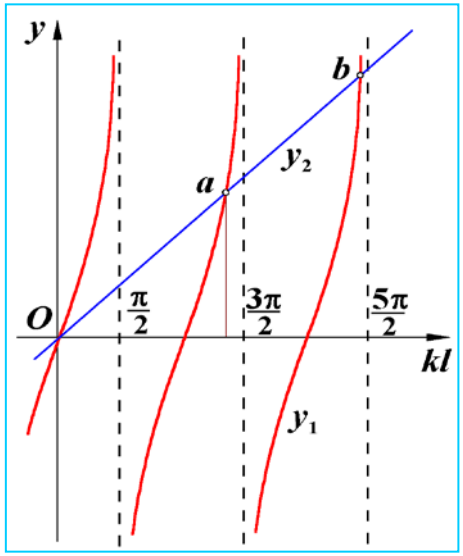
$$(kl)_a = 4.493$$

$$\sqrt{\frac{F}{EI}}\, l = 4.493$$

$$F_{cr} = \frac{4.493^2 EI}{l^2} \approx \frac{\pi^2 EI}{(0.7l)^2}$$

$$y_1 = \tan kl$$

$$y_2 = kl$$

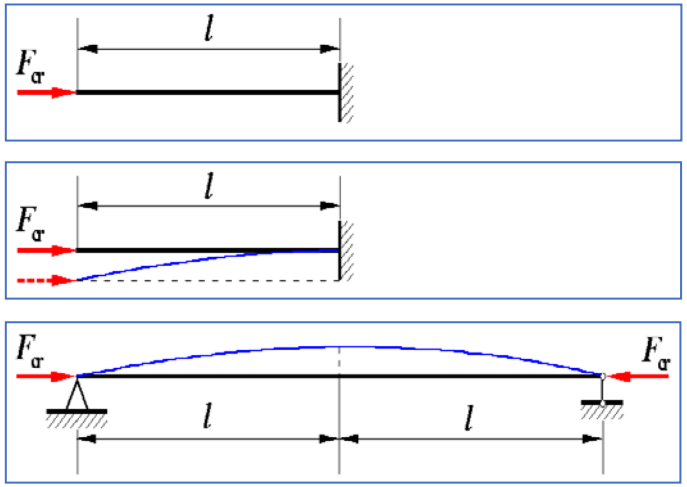


2、欧拉公式的一般形式

类比法确定临界载荷

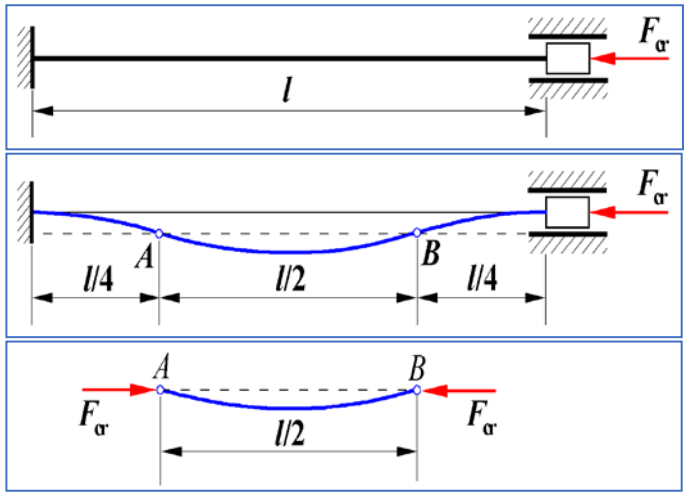
微弯一致，轴压相同

一端固定一端自由



$$F_{cr}=\frac{\pi^2 EI}{(2l)^2}=\frac{\pi^2 EI}{4l^2}$$

两端固定



$$F_{cr}=\frac{\pi^2 EI}{(0.5l)^2}=\frac{4\pi^2 EI}{l^2}$$

2、欧拉公式的一般形式

细长理想直杆临界载荷

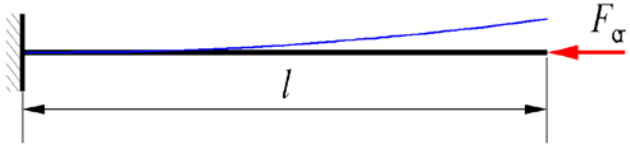
$$F_{cr}=\frac{\pi^2 EI}{(\mu l)^2}$$

μ —长度因数

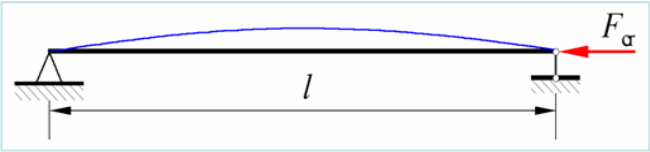
代表支持方式对临界载荷的影响

μl —相当长度

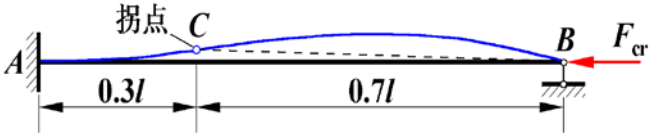
相当于两端铰支细长压杆的长度



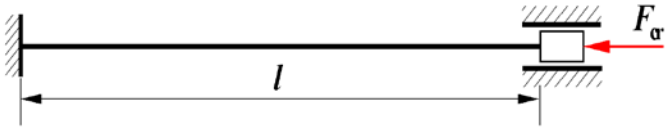
$\mu=2$



$\mu=1$



$\mu=0.7$

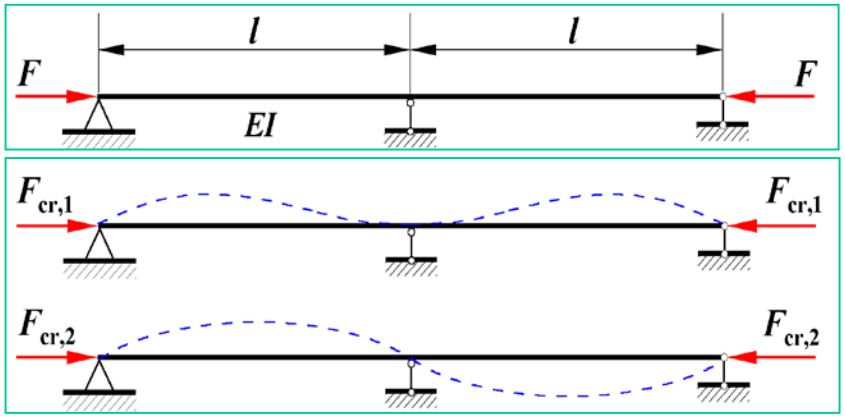


$\mu=0.5$

增强约束，抑制失稳

3、例题

例11-2 等截面细长杆，用类比法求 F_{cr}.



解：1）失稳模态分析 可能的微弯形式

存在对称与反对称两类微弯平衡形式

2）临界载荷计算

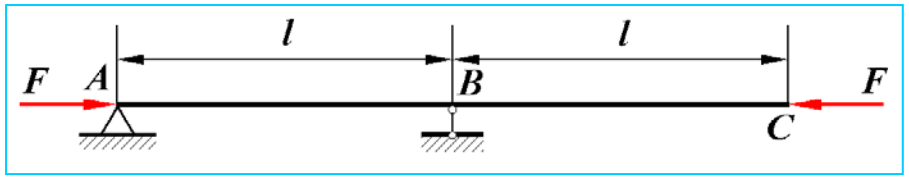
$$l_{eq,1} = 0.7l \qquad F_{cr,1} = \frac{\pi^2 EI}{(0.7l)^2}$$

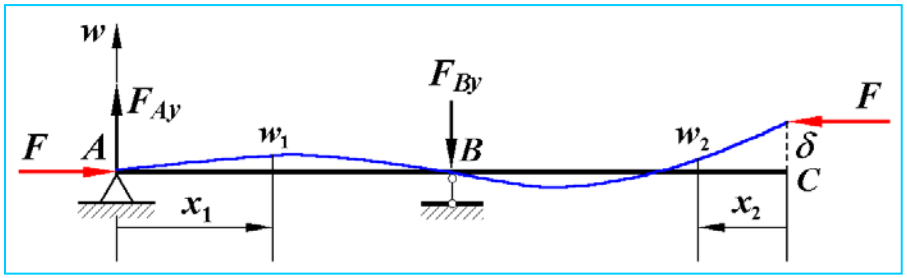
$$l_{eq,2} = l \qquad F_{cr,2} = \frac{\pi^2 EI}{l^2}$$

$$\therefore\ F_{cr} = F_{cr,2} = \frac{\pi^2 EI}{l^2}$$

3、例题

例11-3 图示等截面细长杆，求 F_{cr}.





$w_1, w_2, M(x_1), M(x_1)$均设为正

解： $F_{Ay} = F_{By} = \dfrac{F\delta}{l}$

$$M(x_1) = -Fw_1 + \frac{F\delta}{l}x_1 \qquad M(x_2) = F(\delta - w_2)$$

$$EIw_1'' + Fw_1 = \frac{F\delta}{l}x_1$$

$$w_1 = A_1 \sin kx_1 + B_1 \cos kx_1 + \frac{\delta}{l}x_1$$

$$EIw_2'' + Fw_2 = F\delta$$

$$w_2 = A_2 \sin kx_2 + B_2 \cos kx_2 + \delta$$

$$k^2 = \frac{F}{EI}$$

$x_1 = 0$， $w_1 = 0$

$x_2 = 0$， $w_2 = \delta$ $\Rightarrow$ $B_1 = B_2 = 0$

$x_1 = l$， $w_1 = 0$

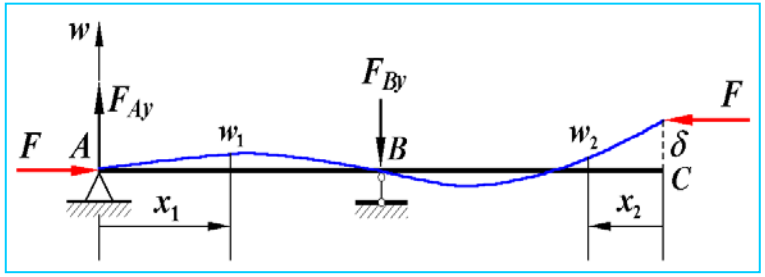
$x_1 = x_2 = l$， $w_1 = w_2$

$x_1 = x_2 = l$， $w'_1 = -w'_2$

3、例题

$$w_1 = A_1 \sin kx_1 + \frac{\delta}{l} x_1$$

$$w_2 = A_2 \sin kx_2 + \delta$$



$$A_1 \sin kl + \delta = 0$$

$$A_1 \sin kl - A_2 \sin kl = 0$$

$$A_1 k \cos kl + A_2 k \cos kl + \frac{\delta}{l} = 0$$

$$\begin{vmatrix} \sin kl & 0 & 1 \\ \sin kl & -\sin kl & 0 \\ k\cos kl & k\cos kl & 1/l \end{vmatrix} = 0$$

$$\boxed{\sin kl\left(\sin kl - 2kl\cos kl\right) = 0}$$

1) $\sin kl = 0$

$$kl = n\pi, \quad F_{cr,1} = \frac{\pi^2 EI}{l^2}$$

$$\delta = 0, \quad A_1 = -A_2$$

$$w_1 = A_1 \sin\frac{\pi}{l} x_1, \quad w_2 = -A_1 \sin\frac{\pi}{l} x_2$$

2) $\sin kl - 2kl\cos kl = 0$

$$F_{cr,2} = \frac{1.34EI}{l^2} = \frac{\pi^2 EI}{(2.71l)^2}$$

$$\boxed{F_{cr} = F_{cr,2} = \frac{1.34EI}{l^2}}$$

本讲：稳定概念与欧拉公式

压杆微弯平衡

下讲：临界应力总图与稳定计算